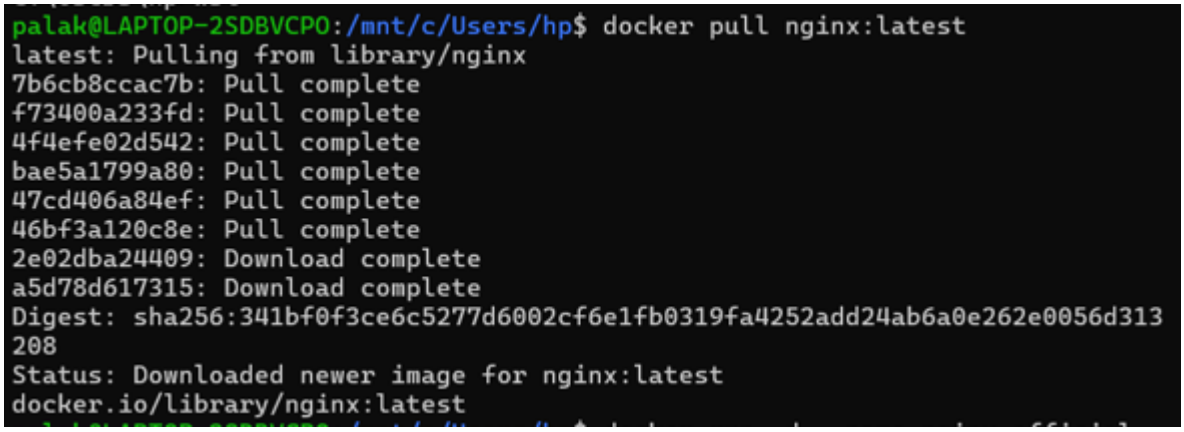
Cloud-containerization-and-DevOps

Experiment 3:- Docker Deploy NGINX Using Different Base Images and Comparing Image Layers

Part 1: Deploy NGINX Using Official Image (Recommended Approach)

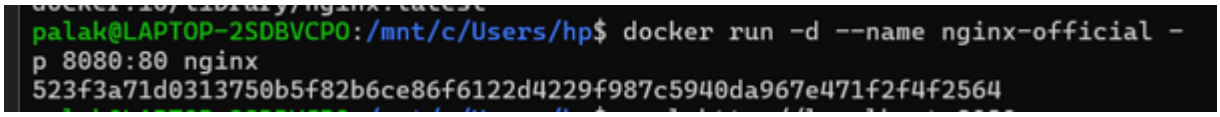
Step 1: Pull the Image

```
docker pull nginx:latest
```



Step 2: Run the Container

```
docker run -d --name nginx-official -p 8080:80 nginx
```



Step 3: Verify

```
curl http://localhost:8080
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>
```

You should see the **NGINX welcome page**.

Key Observations

```
docker images nginx
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker images nginx
REPOSITORY   TAG       IMAGE ID       CREATED      SIZE
nginx        latest    341bf0f3ce6c   2 days ago   240MB
```

- Image is **pre-optimized**
- Minimal configuration required
- Uses Debian-based OS internally

Part 2: Custom NGINX Using Ubuntu Base Image

Step 1: Create Dockerfile

```
FROM ubuntu:22.04

RUN apt-get update && \
    apt-get install -y nginx && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

```
docker build -t nginx-ubuntu .
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/nginx-ubuntu$ docker build -t nginx-ub
untu .
[+] Building 95.0s (6/6) FINISHED                          docker:default
 => [internal] load build definition from Dockerfile                  0.1s
 => => transferring dockerfile: 218B                                  0.0s
 => [internal] load metadata for docker.io/library/ubuntu:22.04       7.1s
 => [auth] library/ubuntu:pull token for registry-1.docker.io         0.0s
 => [internal] load .dockerignore                                     0.1s
 => => transferring context: 2B                                       0.0s
 => [1/2] FROM docker.io/library/ubuntu:22.04@sha256:c7eb020043d8fc2  21.4s
 => => resolve docker.io/library/ubuntu:22.04@sha256:c7eb020043d8fc2a  0.1s
 => => sha256:6f4ebca3e823b18dac366f72e537b1772bc3 29.54MB / 29.54MB  16.8s
 => => extracting sha256:6f4ebca3e823b18dac366f72e537b1772bc3522a5c7a  4.4s
 => ERROR [2/2] RUN apt-get update &&     apt-get install -y nginx &  65.8s
------
 > [2/2] RUN apt-get update &&     apt-get install -y nginx &&     apt-get c
lean &&     rm -rf /var/lib/apt/lists/*:
3.133 Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
3.322 Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129
kB]
4.984 Get:3 http://security.ubuntu.com/ubuntu jammy-security/universe amd64
Packages [1297 kB]
4.996 Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB
]
5.760 Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127
kB]
7.083 Get:6 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages
 [266 kB]
```

Step 3: Run Container

```
docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
```

```
ak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_latest$ docker run --name nginx_latest_container -p 8081:80 -d nginx_latest_custom
ak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_latest$
```

Observations

```
docker images nginx-ubuntu
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker images nginx
REPOSITORY   TAG       IMAGE ID       CREATED      SIZE
nginx        latest    341bf0f3ce6c   2 days ago   240MB
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker build -t nginx-ubuntu .
```

- Much **larger image size**
- More layers
- Full OS utilities available

Part 3: Custom NGINX Using Alpine Base Image

Step 1: Create Dockerfile

```
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

```
docker build -t nginx-alpine .
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/nginx_alpine$ docker build -t nginx_alpine_custom
[+] Building 11.4s (7/7) FINISHED                                docker:default
 => [internal] load build definition from Dockerfile                          0.1s
 => => transferring dockerfile: 104B                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine               1.4s
 => [internal] load .dockerignore                                             0.1s
 => => transferring context: 2B                                               0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb0         8.5s
 => => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb0         0.1s
 => => sha256:955a8478f9aceb66cbf2f579fa3c24e1af278d1 1.21kB / 1.21kB         1.4s
 => => sha256:6d397a54a185dd0b6638d1a3934b81daef7a140 1.40kB / 1.40kB         1.1s
 => => sha256:5e7756927bef33a266e1221356d5da8553139 20.24MB / 20.24MB         5.5s
 => => sha256:6b7b6c7061b76cdb8601e18722d12ae3232f0ddcfa1 954B / 954B         1.4s
 => => sha256:399d0898a94e0084f81499a3e3c29824357118c7ce5 404B / 404B         0.7s
 => => sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15 626B / 626B         0.7s
 => => sha256:bca5d04786e112d958f100a66f8257b2aeefc14 1.86MB / 1.86MB         1.2s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5 3.86MB / 3.86MB         1.9s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6e         0.9s
 => => extracting sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d8         0.9s
 => => extracting sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15d         0.1s
 => => extracting sha256:6b7b6c7061b76cdb8601e18722d12ae3232f0ddcfa1d         0.0s
 => => extracting sha256:399d0898a94e0084f81499a3e3c29824357118c7ce55         0.0s
 => => extracting sha256:955a8478f9aceb66cbf2f579fa3c24e1af278d1fa3ff         0.0s
 => => extracting sha256:6d397a54a185dd0b6638d1a3934b81daef7a140741e1         0.1s
 => => extracting sha256:5e7756927bef33a266e1221356d5da8553139cb80bc5         2.1s
 => [internal] load build context                                             0.1s
 => => transferring context: 77B                                              0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                    0.3s
 => exporting to image                                                        0.7s
 => => exporting layers                                                       0.2s
 => => exporting manifest sha256:3f087e4537f72bfe7cdfe26db640adda44d9         0.0s
 => => exporting config sha256:a9554a547c9f0d96ff56d09998a8879f90a696         0.0s
 => => exporting attestation manifest sha256:c4e4e4f8a5e07f3df03e028a         0.1s
 => => exporting manifest list sha256:e1920b54b1d3050b3ccab60aa8786fd         0.0s
 => => naming to docker.io/library/nginx_alpine_custom:latest                 0.0s
 => => unpacking to docker.io/library/nginx_alpine_custom:latest              0.1s
```

Step 3: Run Container

```
docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/n
ginx_alpine$ docker run --name nginx_alpine_container -p 8083:80 -d nginx_al
pine_custom
ddc95792741066e8482236b372793aa3bd9df37cccba6b38c225a893eb4e5087
```

Observations

```
docker images nginx-alpine
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/n
ginx_alpine$ docker images
REPOSITORY            TAG       IMAGE ID       CREATED          SIZE
nginx_alpine_custom   latest    e1920b54b1d3   23 seconds ago   92.5MB
nginx_ubuntu_custom   latest    bd03b4cee3df   3 days ago       226MB
nginx_latest_custom   latest    60cfe9a9a0c0   3 days ago       237MB
myimage               1.0       6294b3435f74   9 days ago       183MB
mysql                 latest    db32c8ec843c   10 days ago      1.27GB
nginx                 latest    341bf0f3ce6c   11 days ago      240MB
ubuntu                latest    cd1dba651b30   4 weeks ago      119MB
```

- Extremely **small image**
- Fewer packages
- Faster pull and startup time

Part 4: Image Size and Layer Comparison

Compare Sizes

```
docker images | grep nginx
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/n
ginx_alpine$ docker images | grep nginx
nginx_alpine_custom   latest    e1920b54b1d3   5 minutes ago   92.5MB
nginx_ubuntu_custom   latest    bd03b4cee3df   3 days ago      226MB
nginx_latest_custom   latest    60cfe9a9a0c0   3 days ago      237MB
nginx                 latest    341bf0f3ce6c   11 days ago     240MB
```

Typical result (approx):

Image Type	Size
nginx:latest	237 MB
nginx-ubuntu	226 MB
nginx-alpine	92 MB

Inspect Layers

```
docker history nginx
docker history nginx-ubuntu
docker history nginx-alpine
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/nginx_alpine$ docker history nginx_ubuntu_custom
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
bd03b4cee3df   3 days ago    CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      3 days ago    EXPOSE [80/tcp]                                 0B        buildkit.dockerfile.v0
<missing>      3 days ago    COPY index.html /var/www/html/index.html # b…   20.5kB    buildkit.dockerfile.v0
<missing>      3 days ago    RUN /bin/sh -c apt-get update && apt-get ins…   65.9MB    buildkit.dockerfile.v0
<missing>      4 weeks ago   /bin/sh -c #(nop)  CMD ["/bin/bash"]            0B
<missing>      4 weeks ago   /bin/sh -c #(nop) ADD file:3077ee44db3cc7d38…   87.6MB
<missing>      4 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      4 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      4 weeks ago   /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH     0B
<missing>      4 weeks ago   /bin/sh -c #(nop)  ARG RELEASE                  0B
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/nginx_alpine$ docker history nginx_latest_custom
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
60cfe9a9a0c0   3 days ago    COPY index.html /usr/share/nginx/html/index.…   24.6kB    buildkit.dockerfile.v0
<missing>      11 days ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      11 days ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      11 days ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      11 days ago   COPY 30-tune-worker-processes.sh /docker-ent…   16.4kB    buildkit.dockerfile.v0
<missing>      11 days ago   COPY 20-envsubst-on-templates.sh /docker-ent…   12.3kB    buildkit.dockerfile.v0
<missing>      11 days ago   COPY 15-local-resolvers.envsh /docker-entryp…   12.3kB    buildkit.dockerfile.v0
<missing>      11 days ago   COPY 10-listen-on-ipv6-by-default.sh /docker…   12.3kB    buildkit.dockerfile.v0
<missing>      11 days ago   COPY docker-entrypoint.sh / # buildkit          8.19kB    buildkit.dockerfile.v0
<missing>      11 days ago   RUN /bin/sh -c set -x     && groupadd --syst…   86.6MB    buildkit.dockerfile.v0
<missing>      11 days ago   ENV DYNPKG_RELEASE=1~trixie                     0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENV PKG_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENV ACME_VERSION=0.3.1                          0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENV NJS_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENV NJS_VERSION=0.9.5                           0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENV NGINX_VERSION=1.29.5                        0B        buildkit.dockerfile.v0
<missing>      11 days ago   LABEL maintainer=NGINX Docker Maintainers <d…   0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/nginx_ubuntu/nginx_alpine$ docker history nginx
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
341bf0f3ce6c   11 days ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      11 days ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      11 days ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      11 days ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      11 days ago   COPY 30-tune-worker-processes.sh /docker-ent…   16.4kB    buildkit.dockerfile.v0
<missing>      11 days ago   COPY 20-envsubst-on-templates.sh /docker-ent…   12.3kB    buildkit.dockerfile.v0
```

Observations:

- Ubuntu has many filesystem layers
- Alpine has minimal layers
- Official NGINX image is optimized but heavier than Alpine

Part 5: Functional Tasks Using NGINX

Task 1: Serve Custom HTML Page

```
mkdir html
echo "<h1>Hello from Docker NGINX</h1>" > html/index.html
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment$ mkdir html
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment$ echo "<h1>Hello from Docker NGINX</h1>" > html/index.html
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment$ nano Dockerfile
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment$ ls
Dockerfile  html  nginx_latest  nginx_ubuntu
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment$ cd html/
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/html$ ls
index.html
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/docker-nginx-experiment/html$
```

Run:

```
docker run -d \
  -p 8083:80 \
```

```
    -v $(pwd)/html:/usr/share/nginx/html \
    nginx
```

Task 2: Reverse Proxy (Conceptual)

NGINX can:

- Forward traffic to backend services
- Load balance multiple containers
- Terminate SSL

Example use cases:

- Frontend for microservices
- API gateway
- Static file server

Part 6: Comparison Summary

Image Comparison Table

Feature	Official NGINX	Ubuntu + NGINX	Alpine + NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

Part 7: When to Use What

Use Official NGINX Image When:

- Production deployment
- Standard web hosting
- Reverse proxy / load balancer

Use Ubuntu-Based Image When:

- Learning Linux + NGINX internals
- Heavy debugging needed
- Custom system-level dependencies

Use Alpine-Based Image When:

- Microservices
- CI/CD pipelines
- Cloud and Kubernetes workloads